



SPECIAL ISSUE RESEARCH ARTICLE

Theoretical study on the catalytic mechanism of oxyluciferin hydrolysis by luciferin-regenerating enzyme in firefly bioluminescence substrate regeneration

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Abstract

Firefly bioluminescence (BL) is widely utilized in bioimaging due to its high quantum yield; however, its performance in long-term monitoring is limited by rapid signal decay resulting from product inhibition by oxyluciferin (**oLu**). In nature, fireflies employ the luciferin-regenerating enzyme (LRE) to catalyze **oLu** hydrolysis, thereby removing inhibition and initiating substrate regeneration for subsequent emission. While previous studies have established the three-step regeneration pathway—LRE-catalyzed **oLu** hydrolysis, CHBT condensation with L-Cys, and L/D-luciferin isomerization—the initiating step lacks molecular-level clarity. Specifically, how the active site environment of LRE facilitates this reaction remains unknown, which hinders a comprehensive understanding of the recycling process and the rational optimization of sustained bioluminescent systems. To address this, we investigate its microscopic mechanism using molecular dynamics (MD) simulations and quantum mechanics/molecular mechanics (QM/MM) calculations. The results reveal that LRE provides a critical residue, Asp111, which acts as a general base to facilitate nucleophilic addition. The subsequent ring opening follows a stepwise mechanism, characterized by sequential C4-N3 bond cleavage and proton transfer-coupled C2-S1 bond breaking as the rate-determining step. Overall, this work elucidates the molecular role of LRE in substrate regeneration and provides a theoretical foundation for understanding how fireflies restart their bioluminescent cycles.

KEYWORDS

firefly bioluminescence, luciferin-regenerating enzyme, oxyluciferin hydrolysis, reaction mechanism, substrate regeneration

Abbreviations: ANL, adenylating enzyme superfamily; Asp, aspartate; ATP, adenosine triphosphate; BL, bioluminescence; CHBT, 2-cyano-6-hydroxybenzothiazole; CoA, coenzyme A; Cys, cysteine; GAFF, General Amber Force Field; INT, intermediate; L-Cys, L-cysteine; LRE, luciferin-regenerating enzyme; MD, molecular dynamics; MM-PBSA, molecular mechanics Poisson–Boltzmann surface area; NPT, isothermal-isobaric ensemble; NVT, canonical ensemble; oLu, oxyluciferin; PDB, Protein Data Bank; PES, potential energy surface; PME, particle mesh Ewald; QM/MM, quantum mechanics/molecular mechanics; RESP, restrained electrostatic potential; RMSD, root-mean-square deviation; TS, transition state; WAT, water molecule.

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INTRODUCTION

Firefly bioluminescence (BL) is widely applied to *in vivo* imaging^{1,2} and analytical detection^{3,4} due to its high quantum yield.⁵ To date, considerable research efforts have focused on enhancing its performance, primarily by regulating emission spectra for deep-tissue penetration^{6–9} and utilizing environmental sensitivity for bio-sensing.¹⁰ However, in practical applications, firefly BL signals often exhibit rapid decay after an initial burst,¹¹ and this instability remains a major limitation for its further development. This signal decay has been attributed to product inhibition,¹² a phenomenon where the BL reaction byproduct oxyluciferin (**oLu**) has extremely strong affinity for the luciferase active center. After photon release, **oLu** remains firmly retained in the enzyme's active center,¹³ blocking the binding of substrate D-luciferin and thus inhibiting subsequent luminescence reactions.

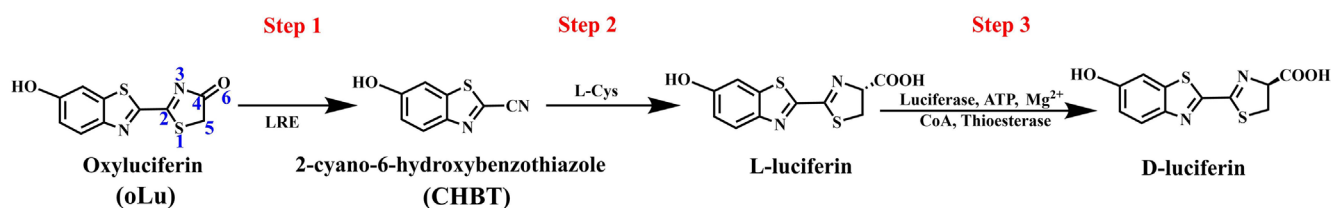
This product inhibition effect seems to contradict the physiological fact that adult fireflies barely eat yet can maintain high-frequency flashing.¹⁴ Regarding the limitations brought by substrate consumption and product accumulation after a single luminescence event, how fireflies achieve subsequent luminescence has become a research focus. Researchers speculated that there might exist an efficient substrate regeneration mechanism in their bodies, namely recycling the retained byproduct **oLu** and further converting it into D-luciferin to form a closed-loop material cycle, thereby ensuring continuous substrate supply and cyclic luminescence reactions.¹³

This hypothesis has been supported by multiple experimental studies.^{15–18} In 1974, Okada et al. injected ¹⁴C-labeled **oLu** into fireflies and successfully detected labeled D-luciferin¹⁶; in 1976, McCapra et al. further discovered that ¹⁴C-labeled cysteine (Cys) could be integrated into the regenerated luciferin skeleton.¹⁷ In 2001, Gomi et al. first identified and expressed luciferin-regenerating enzyme (LRE) and also confirmed that under its catalysis, **oLu** rapidly converts to the intermediate 2-cyano-6-hydroxybenzothiazole (CHBT).¹⁵ These studies together revealed the pathway where **oLu** first converts to CHBT and then reacts with Cys to generate D-luciferin.

However, this regeneration pathway still faces a critical challenge at the chirality level: luminescence reactions specifically consume D-luciferin, but amino acids in organisms mostly exist in L-type conformations,¹⁹ suggesting that the primary product of the cycle may be L-luciferin without luminescence activity, followed by chirality inversion. Regarding the specific mechanism of this chirality inversion, in 2005, Nakamura et al. discovered through enzymatic experiments that luciferase is a bifunctional enzyme that not only catalyzes luminescence but also possesses acyl-CoA synthetase activity.²⁰ This functional diversity stems from its structural membership in the adenylating enzyme ANL superfamily, showing significant homology with CoA synthetase.²¹ Experimental results further confirmed that with the synergistic participation of ATP, Mg²⁺, CoA, and esterase, luciferase can drive the stereoisomerization of L-luciferin.²⁰ Subsequently, in 2006, Niwa et al. further confirmed through *in vivo* component detection that D-Cys barely exists in fireflies,¹⁹ thus confirming at the physiological level that the D-type substrate required for luminescence is actually transformed from the L-type precursor through the enzymatic isomerization pathway.

Based on the cumulative findings described above, a general framework for the firefly D-luciferin regeneration mechanism has been established. As proposed by Cheng et al., the complete regeneration process comprises three consecutive steps,²² as shown in Scheme 1: (i) the LRE-catalyzed hydrolysis of **oLu** to generate CHBT;^{15,23} (ii) the condensation of CHBT with L-Cys to generate L-luciferin;^{15,24} and (iii) the stereo isomerization of L-luciferin to D-luciferin.²⁵ In this pathway, the second step is characterized by a high spontaneous tendency and proceeds without enzymatic participation;^{26,27} its microscopic mechanism has been previously elucidated through theoretical simulations in an aqueous environment.²² Regarding the third step, the identification of the key racemase (e.g., AteACOT1) and subsequent enzymatic experiments have clarified the biochemical route of stereoisomerization at the molecular level,²⁸ making the understanding of subsequent steps relatively complete.

In contrast, the first step—the LRE-catalyzed hydrolysis of **oLu**—represents the initiating event of the entire cycle, directly responsible for the removal of product inhibition



SCHEME 1 Schematic representation of firefly luciferin regeneration and numbering scheme for the thiazole ring of oxyluciferin (**oLu**).

and the efficient restart of the BL system. Although Cheng et al. provided a preliminary exploration of this reaction in an aqueous solution,²² the specific catalytic mechanism within the LRE enzyme environment remains elusive, primarily due to the current lack of a high-resolution complex structure of LRE bound with its substrate. Given that the enzyme's active site environment can significantly modulate chemical reactivity, a detailed investigation into the catalytic process within the enzyme is required. To address this, theoretical calculations were employed to elucidate the microscopic mechanism of LRE-catalyzed **oLu** hydrolysis in this paper, aiming to reveal the molecular basis of firefly substrate recycling.

MATERIALS AND METHODS

Computational methods

Setup of systems

The simulation system of the LRE-**oLu** complex was constructed based on the crystal structure of wild-type LRE from *P. pyralis* (PDB ID: 5GTQ).²⁹ During protein preparation, hydrogen atoms were added at pH 8.0 using the H++ web server to mimic the weakly alkaline in vivo environment.^{15,30–32} Subsequently, **oLu** was docked into LRE using the Autodock Vina program.³³ The docking grid box was centered at (33.846, 14.809, 17.572) with a box size of $15 \times 15 \times 15 \text{ \AA}^3$, covering the potential active site region. Considering that empirical docking scoring functions are approximations with inherent uncertainties,³⁴ pose selection in computational enzymology often incorporates the consideration of catalytically productive orientations that satisfy the structural requirements for the chemical reaction, rather than relying on ranking by scores alone.^{35,36} In our docking results, as the top-ranked pose lacked a suitable general base in its vicinity to facilitate hydrolysis (see Figure S1), a catalytically productive pose in which Asp111 is appropriately positioned to function as a general base was identified and selected. Then, parameters were set for the complex. The protein was described using the Amber ff19SB force field,³⁷ while **oLu** was parameterized using the General Amber Force Field (GAFF).³⁸ The partial charges of **oLu** were calculated at the HF/6-31G* level using Gaussian 09 software³⁹ and derived through the restrained electrostatic potential (RESP) fitting method.⁴⁰ Missing force field parameters were generated using the parmchk2 tool in AmberTools. Finally, all systems were charge-neutralized by adding Na^+ and explicitly solvated in TIP3P water boxes,⁴¹ with the minimum distance between the water box boundary and protein surface being $12 \text{ \AA}$.

MD simulation of systems

After proper setups, the system was equilibrated in several steps: (1) energy minimization was performed using the steepest descent and conjugate-gradient methods sequentially; (2) the system was then gradually heated from 0 K to 300 K over 400 ps in the NVT ensemble, with weak constraints applied to the protein backbone; (3) density equilibration was conducted for 1 ns at 300 K and 1.0 atm, utilizing a Langevin thermostat⁴² and a Berendsen barostat⁴³ with a collision frequency of 2 ps^{-1} and a pressure relaxation time of 1 ps; (4) a subsequent 3-ns NPT equilibration was carried out to further stabilize the pressure and temperature; (5) a 100-ns MD simulation was conducted. Nonbonded interactions were treated with the Particle Mesh Ewald (PME) method⁴⁴ with a cutoff of $12 \text{ \AA}$. The covalent bonds involving hydrogen atoms were constrained using the SHAKE algorithm,⁴⁵ enabling a 2-fs integration step. All simulations were performed with the Amber21 software.⁴⁶ Based on MD simulation trajectories, the mmpbsa.py program was used to calculate the binding free energy of **oLu** with LRE and the contribution of residues around **oLu** to the binding free energy using the MM-PBSA method.⁴⁷ For comparison, the binding free energy of **oLu** to firefly luciferase (PDB ID: 4G36)⁴⁸ was also calculated using the same MM-PBSA method based on its corresponding MD simulation trajectories.

QM/MM calculations

QM/MM calculations were performed based on configurations after MD equilibration. The reaction center, namely, **oLu**, the water molecule WAT participating in the hydrolysis reaction, and the key residue Asp111 were included in the QM region, while the remaining atoms were assigned to the MM region. Specifically, for the computational model **oLu + WAT** mentioned below, the QM region includes **oLu** and one water molecule, WAT, participating in the reaction, with the remaining atoms as the MM region. For the computational model **oLu + WAT + Asp**, the QM region includes **oLu**, one water molecule WAT, and Asp111, with the remaining atoms as the MM region. For other computational models, **oLu + 2WAT**, **oLu + 3WAT**, and **oLu + 2WAT + Asp**, the treatment is similar. All systems used the electrostatic embedding scheme to handle electrostatic interactions between QM and MM regions.⁴⁹ Geometry optimization and potential energy surface (PES) scans of the QM region were performed at the B3LYP-D3/6-31G* level, with the MM region treated using the Amber ff19SB force field.³⁷ Subsequently, based on structures optimized at the B3LYP-D3/6-31G*/MM

level, single-point energy corrections were performed on the QM region at the M06-2X/def2-TZVP computational level. This dual-level strategy (M06-2X//B3LYP-D3) has been widely validated for its reliability in investigating enzyme-catalyzed reaction mechanisms,⁵⁰ offering an effective balance between structural accuracy and energetic precision. Atoms within 5 Å of **oLu** in the MM region were allowed to relax, while the remaining atoms were fixed. All QM/MM calculations were completed using ChemShell software,^{51,52} calling ORCA⁵³ and DL_POLY⁵⁴ for QM and MM region calculations, respectively.

QM/MM MD simulation

To compare the reaction conformations of the **oLu** + **WAT** + **Asp** and **oLu** + **2WAT** + **Asp** models, 20 ps QM/MM MD simulations at 300 K and 1 atm with a time step of 2 fs were performed. For the **oLu** + **WAT** + **Asp** model, the QM region includes **oLu**, Asp111, and 1 water molecule; for the **oLu** + **2WAT** + **Asp** model, the QM region includes **oLu**, Asp111, and 2 water molecules; the remaining parts were treated as the MM region. The QM region was treated at the B3LYP-D3/6-31G* level, while the MM region was treated using the Amber ff19SB force field.³⁷ Chemical bonds crossing regions were truncated and saturated with hydrogen link atoms, and interactions between the two regions were treated using the electrostatic embedding scheme. QM calculations were completed using the TeraChem program package,⁵⁵ and MM calculations and QM/MM MD simulations were completed using the Amber21 program package.⁴⁶

RESULTS AND DISCUSSION

Structural dynamics and proposed catalytic mechanisms of the LRE-**oLu** complex

As shown in Figure S2, during the 100 ns MD simulation, the root-mean-square deviation (RMSD) of the protein backbone and that of the substrate **oLu** both remained stable after an initial period. This high structural stability is complemented by a robust total binding free energy of −15.4 kcal/mol, reflecting a strong binding affinity between LRE and **oLu** and ensuring a reliable structural basis for investigating the catalytic mechanism. Notably, the binding free energy of **oLu** to firefly luciferase is calculated to be −19.9 kcal/mol (Figure S3), which is lower than that of **oLu** to LRE. This stronger affinity confirms the potent product inhibition effect mentioned in the **Introduction** and indicates that the alleviation of this inhibition is likely a dynamic

process rather than a static competition. By continuously hydrolyzing **oLu** molecules dissociated from luciferase,⁵⁶ LRE effectively prevents them from re-binding, thereby driving the steady release of the **oLu** trapped within the luciferase active site and ensuring the overall feasibility of the substrate regeneration pathway. Based on contact frequencies and per-residue energy decompositions (Figure 1A,B), five key residues—Glu18, Asp111, Lys113, Phe259, and Phe301—were found to interact closely with the substrate. Notably, while Glu18 maintained a proximity frequency of approximately 40% within 5 Å of **oLu**, the other four residues exhibited nearly 100% occupancy, reflecting their exceptional stability within the binding pocket during the simulation. The overall contribution of these five residues to binding energy and the contribution of electrostatic interactions are shown in Figure 1C. It is indicated that the total contribution is mainly electrostatic. Specifically, Lys113, Phe259, and Phe301 contribute favorably to binding, with the latter two stabilizing the benzothiazole side of **oLu** through hydrogen bond networks (Figure 1D). In contrast, Glu18 and Asp111 exhibit unfavorable binding energies but occupy strategic positions on the thiazole side. Notably, the stably positioned and negatively charged Asp111 remains situated adjacent to the C4 atom—the site for hydrolysis. This consistent spatial proximity validates our initial pose selection and suggests that Asp111 likely functions as a general base to facilitate proton transfer during the subsequent catalytic process.

The occurrence of enzyme-catalyzed hydrolysis depends not only on stable substrate binding but also, critically, on the effective accumulation of nucleophilic reagents (water molecules) at the reaction site. Figure S4 shows the quantity and distance distribution of water molecules around the substrate C4 atom during the 100 ns MD simulation. The results show that throughout the equilibration phase, multiple water molecules remain stably positioned around the C4 atom (Figure S4A), and the shortest distance between water oxygen atoms and the C4 atom is primarily centered between 3.5 and 4.0 Å (Figure S4B). This proximal geometric arrangement provides favorable conditions for nucleophilic attack; furthermore, these water molecules form hydrogen bond networks that can serve as potential pathways for proton transfer.

Based on the above analysis, the catalytic pocket of LRE provides conditions conducive to substrate hydrolysis: the substrate binding conformation is stable, nucleophilic water molecules are optimally positioned near the reaction site, and the potential catalytic base Asp111 is spatially well-oriented for catalysis. From the perspective of chemical mechanism, the hydrolysis of **oLu** typically includes two key elementary reaction stages: first, the nucleophilic addition reaction of water molecules to the C4 atom of substrate **oLu**, forming an addition intermediate; subsequently, the carbon-heteroatom (S or N) bond of the intermediate breaks, completing ring opening. Among

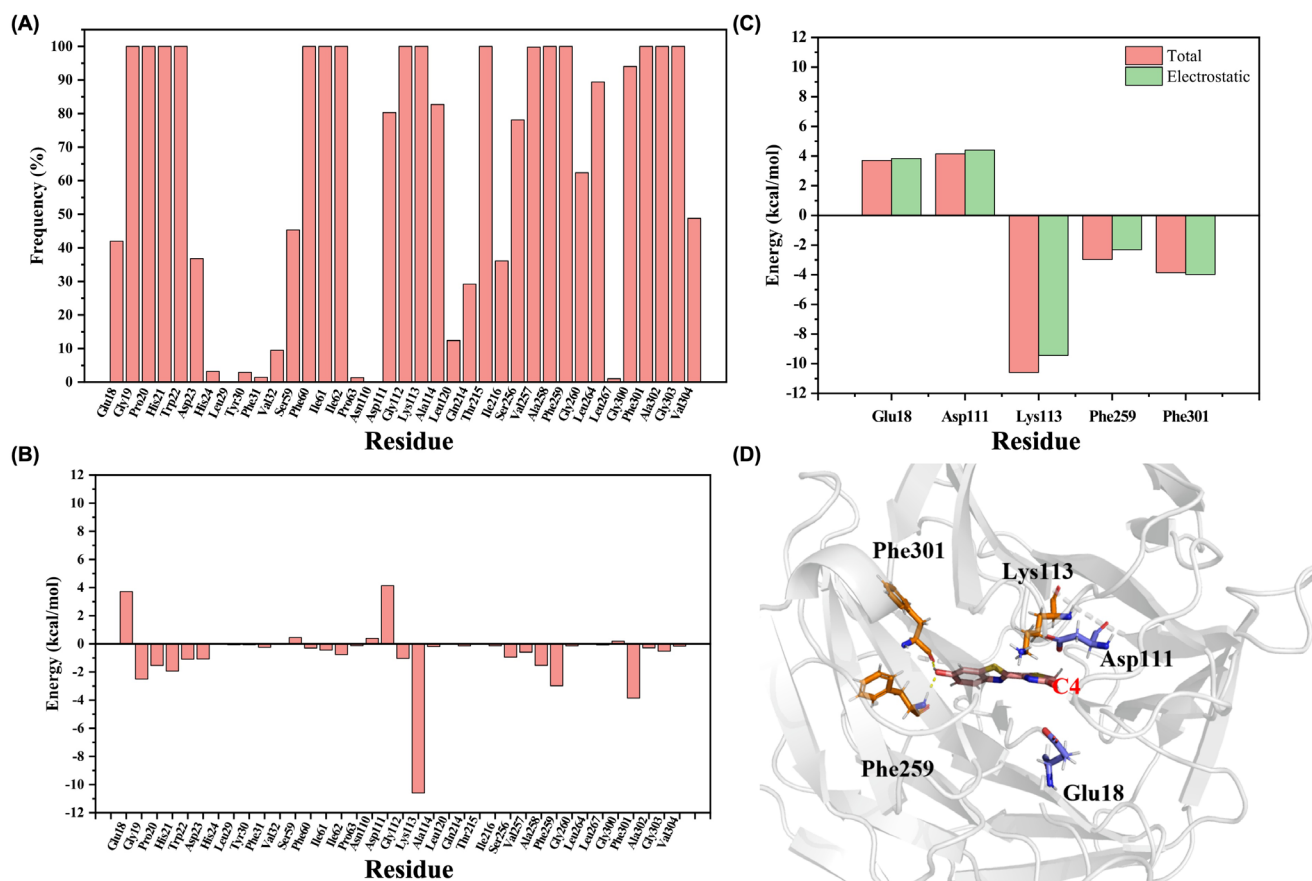


FIGURE 1 Interaction analysis of the LRE-**oLu** complex. (A) Contact frequency of residues within 5 Å of the substrate. (B) Binding free energy contribution per residue. (C) Total and electrostatic energy contributions of key residues. (D) Structural snapshot showing the spatial distribution of **oLu** and the identified key residues in the equilibrated system.

them, the first step of nucleophilic addition is often the key to determining reaction feasibility. Based on the above structural analysis, we proposed two possible pathways for the first-step addition reaction:

Pathway 1 (Non-base-dependent direct nucleophilic addition): Water molecules gathered in the pocket directly initiate nucleophilic attack on the substrate C4 atom, accompanied by proton transfer through the water molecules' hydrogen bond network to induce the addition reaction.

Pathway 2 (Asp111-assisted nucleophilic addition): Nucleophilic water molecules act synergistically with Asp111. Asp111 acts as a general base to abstract a proton from a water molecule, thereby enhancing its nucleophilicity for the subsequent attack on the C4 atom to form the addition intermediate.

Non-base-dependent direct nucleophilic addition pathway

To evaluate the feasibility of the non-base-dependent mechanism, theoretical calculations were first performed

on Pathway 1. Three models (**oLu** + **nWAT**, $n = 1-3$) were constructed to systematically investigate the influence of water cluster size on the addition process, accounting for the dynamic hydrogen bond networks observed in the LRE pocket. The resulting potential energy profiles and key stationary point geometries are shown in Figure 2A,B, respectively.

Regarding the evolution of geometric structures, although all three models undergo C-O bond formation and proton transfer, they exhibit distinct conformational characteristics in their microscopic mechanisms. In the single-water molecule model (**oLu** + **WAT**), a single water molecule mediates both the nucleophilic attack and proton transfer. In the corresponding transition state TS1A, the water molecule simultaneously acts as the nucleophilic reagent and proton donor (C4-Oa: 1.465 Å, O6-Hb: 1.105 Å). This leads to highly synchronized bond formation, bond cleavage, and proton migration, forcing the system into a highly distorted four-membered ring transition state with significant ring strain. Due to this severe conformational constraint, this pathway exhibits the highest activation barrier of 33.3 kcal/mol. In the dual-water model (**oLu** + **2WAT**), the mechanism shifts to a

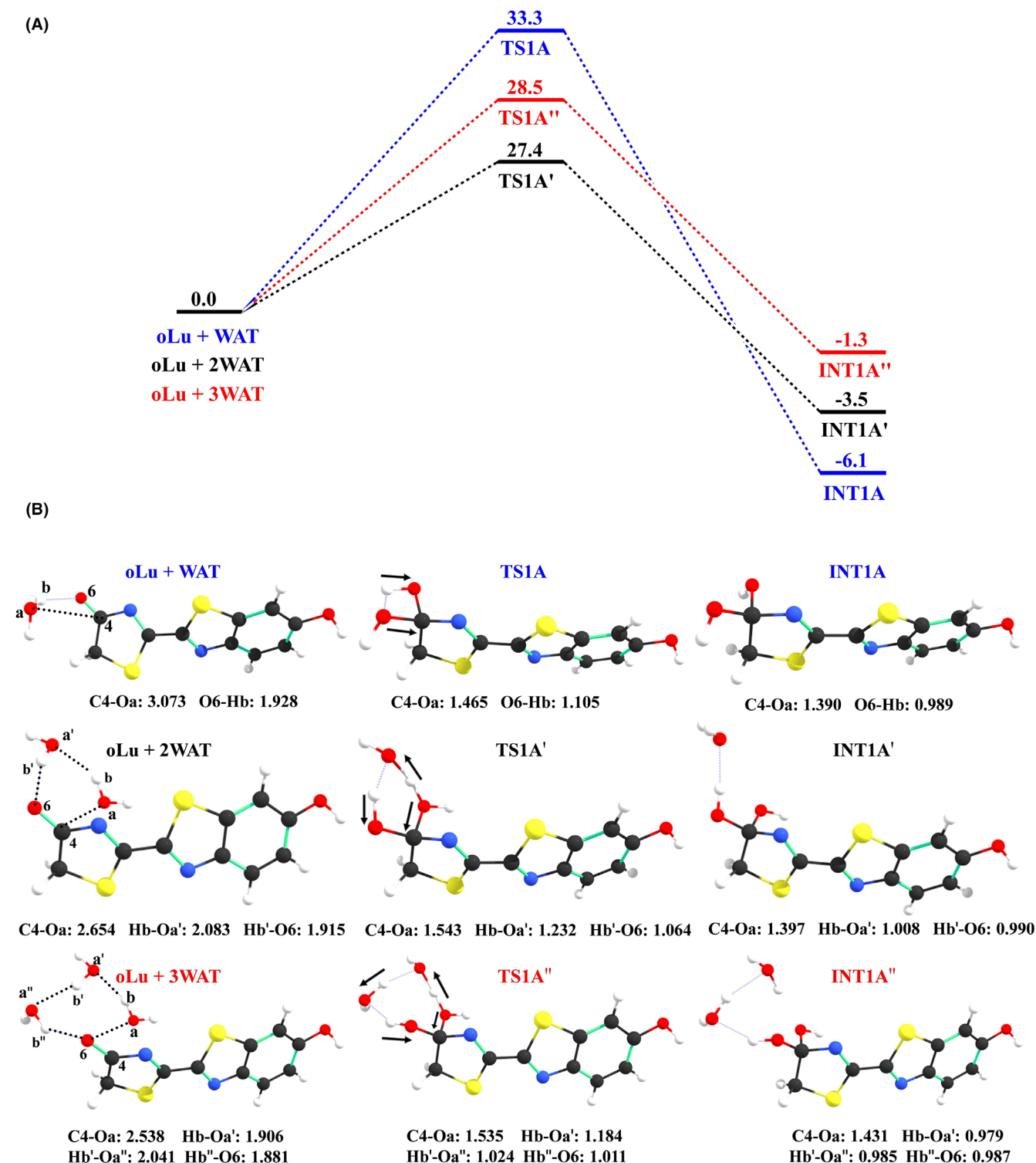


FIGURE 2 (A) Energy profile (ΔE , kcal/mol) and (B) geometric parameters ($\text{\AA}$) of key stationary points for the addition reaction between **oLu** and water of three models (**oLu** + **nWAT**), calculated at the M06-2X/def2-TZVP//B3LYP-D3/6-31G*/MM level of theory. Atoms are colored according to the standard CPK convention: C (black), N (blue), O (red), S (yellow), and H (white).

concerted proton relay. Protons are transferred via a dual-water bridge: one water molecule attacks the C4 atom (C4-Oa: 1.543 $\text{\AA}$) while transferring its proton to a second water molecule (Hb-Oa': 1.232 $\text{\AA}$), which simultaneously shuttles its own proton to the O6 atom (Hb'-O6: 1.064 $\text{\AA}$).

This bridge allows the formation of a more relaxed six-membered ring transition state (**TS1A'**), effectively alleviating geometric strain and reducing the energy barrier by 5.9 kcal/mol to 27.4 kcal/mol—the lowest among the three models. Furthermore, in the triple-water model

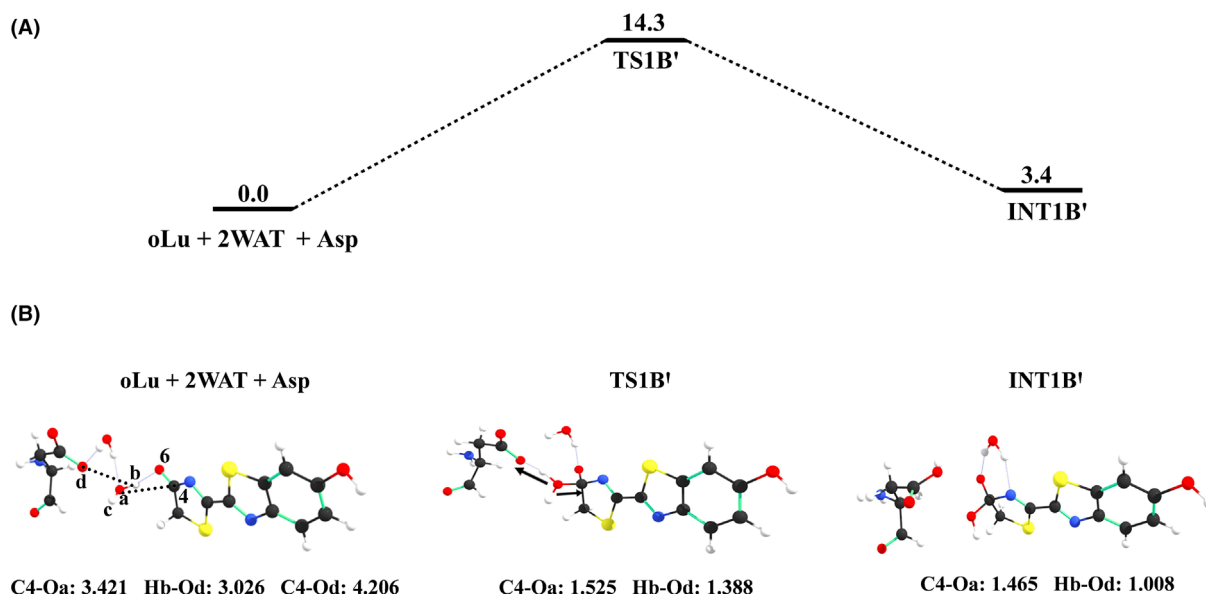


FIGURE 3 (A) Energy profile (ΔE , kcal/mol) and (B) geometric parameters ($\text{\AA}$) of key stationary points for the addition reaction between **oLu** and water of model **oLu + 2WAT + Asp**, calculated at the M06-2X/def2-TZVP//B3LYP-D3/6-31G*/MM level of theory. Atoms are colored according to the standard CPK convention: C (black), N (blue), O (red), S (yellow), and H (white).

(**oLu + 3WAT**), while the three water molecules form a more extended hydrogen-bond chain and an eight-membered ring transition state (**TS1A''**), the energy barrier does not decrease further but instead increases slightly to 28.5 kcal/mol. This suggests that for this system, the six-membered ring achieves an optimal balance between strain relief and efficient proton transfer. In contrast, the eight-membered ring requires a higher degree of spatial organization and conformational synchronization between the water cluster and the substrate, thereby introducing a higher reorganization energy that offsets the stabilization of the transition state. Therefore, based on both structural and energetic analyses, the dual-water concerted relay mechanism is identified as the optimal non-base-dependent pathway.

However, the energy barrier of 27.4 kcal/mol for the **oLu + 2WAT** model is still too high to be overcome under physiological conditions, making this non-base-dependent pathway unfeasible. This indicates that relying solely on water molecules is insufficient for the reaction to occur. Moreover, these findings provide a possible energetic rationale for our initial docking model selection. If **oLu** adopted the top-ranked docking pose, which lacked a nearby general base, the reaction would likely be expected to follow this unfeasible non-base-dependent route, potentially resulting in a lack of catalytic hydrolytic activity. Consequently, the orientation that positions **oLu** near Asp111 appears more consistent with the observed biological function of LRE. As hypothesized, Asp111 is likely required to activate the water molecule and further lower the energy barrier.

Asp111-assisted nucleophilic addition pathway

Based on the results from Pathway 1, which showed that extending the water chain beyond two molecules provided no additional advantage, only the **oLu + WAT + Asp** and **oLu + 2WAT + Asp** models were constructed for Pathway 2. While no feasible transition state was found for the single-water model (Figure S5), a concerted addition pathway was identified for the dual-water system.

The energy profile and stationary point geometries for the **oLu + 2WAT + Asp** model are presented in Figure 3. In the transition state **TS1B'**, the C4-Oa bond is formed (1.525 Å) as Asp111 abstracts a proton (Hb) from the nucleophilic water (Hb-Od = 1.388 Å). In this concerted step, Asp111 acts as a general base to activate the water molecule. The resulting energy barrier is 14.3 kcal/mol, significantly lower than the 27.4 kcal/mol of the non-base-dependent dual water model. This sharp reduction confirms that the participation of Asp111 is the key factor in enabling the hydrolysis reaction.

It is noteworthy that although the **oLu + 2WAT + Asp** model also contains two water molecules, only one water molecule and Asp111 directly participate in the bond formation and cleavage in **TS1B'**. This observation raises the question of why the single-water model (**oLu + WAT + Asp**) is unfeasible, given that it contains the same direct chemical participants. To clarify the role of the second water molecule, the geometric features of both systems were analyzed. As shown in Figure S6, in the single-water model, the nucleophilic attack tendency

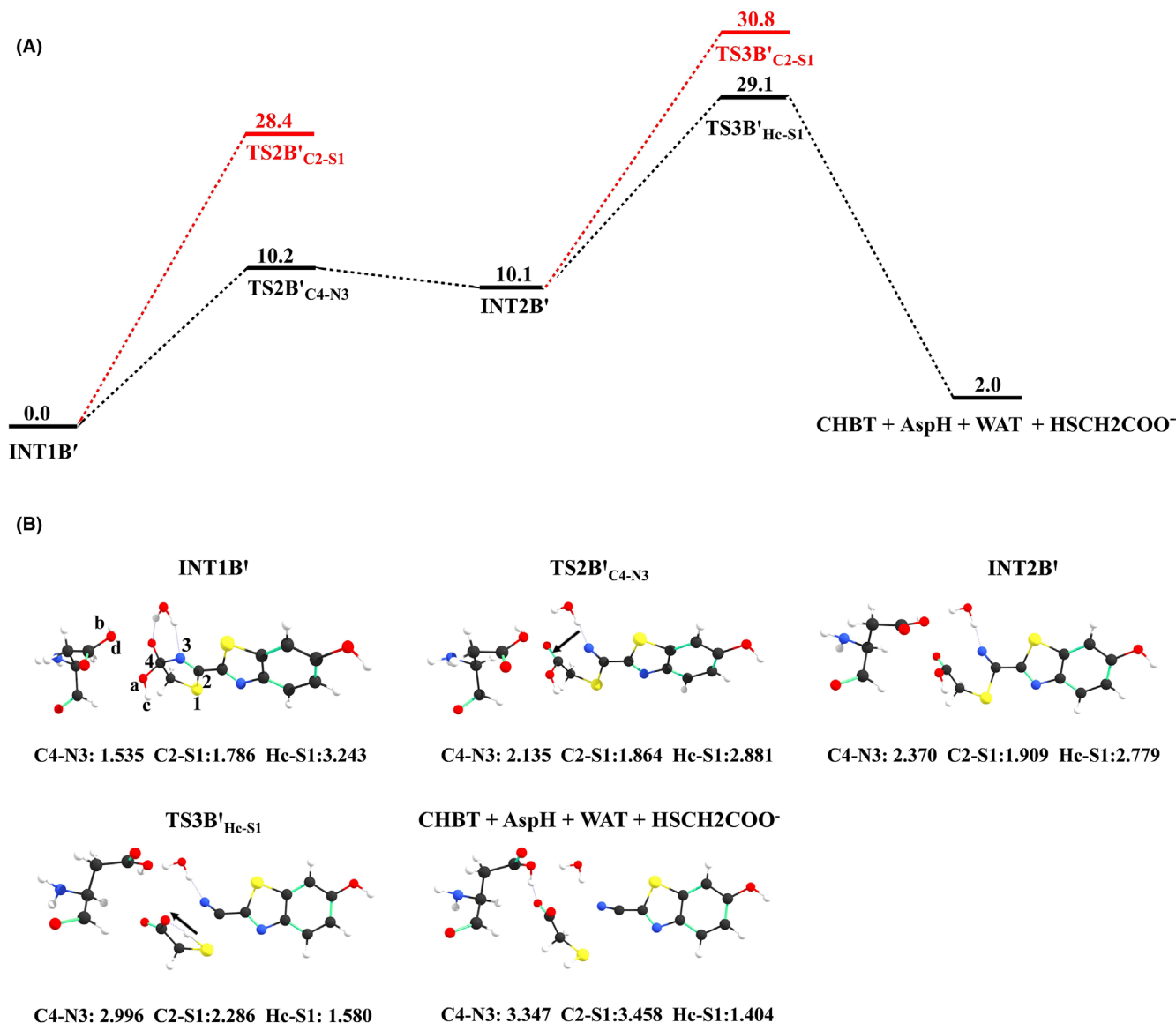


FIGURE 4 (A) Energy profile (ΔE , kcal/mol) and (B) geometric parameters ($\text{\AA}$) of key stationary points for the ring-opening reaction of **INT1B'**, calculated at the M06-2X/def2-TZVP//B3LYP-D3/6-31G*/MM level of theory. Atoms are colored according to the standard CPK convention: C (black), N (blue), O (red), S (yellow), and H (white).

(C-Oa distance) and the base-substrate arrangement (C4-Od distance) are hindered by a strong localized constraint between Asp111 and the lone water molecule. These distances are primarily distributed around 4.25 $\text{\AA}$ and 5.25 $\text{\AA}$, respectively, preventing the system from reaching a reactive state. In contrast, the second water molecule in the **oLu + 2WAT + Asp** model modulates the local environment, thereby enhancing the proton transfer tendency (Hb-Od distance) and facilitating a more compact arrangement. This structural tuning ensures that the nucleophilic water and Asp111 are properly positioned, providing the necessary conditions for the concerted reaction to occur.

In summary, Pathway 2 is identified as the optimal mechanism for the addition stage, forming a stable intermediate **INT1B'** with a barrier of 14.3 kcal/mol. Based on

this intermediate, the subsequent ring-opening process was further investigated.

Ring-opening reaction of **INT1B'**

In intermediate **INT1B'**, the transition of the C4 atom to sp^3 tetrahedral configuration disrupts the planar conjugation of the thiazole ring, rendering it susceptible to opening. This process, which requires the breaking of both the C4-N3 and C2-S1 bonds, is summarized in Figure 4. The calculated energy profiles (Figure 4A) indicate that the reaction proceeds through a stepwise mechanism, initiated by the cleavage of the C4-N3 bond. Breaking the C2-S1 bond at this stage is unfeasible, as it requires overcoming

a high barrier of 28.4 kcal/mol (**TS2B'**_{C2-S1}). In contrast, the barrier for C4-N3 bond cleavage (**TS2B'**_{C4-N3}) is only 10.2 kcal/mol, identifying it as the preferred initiation site. This indicates that the reaction tends to preferentially break the C4-N3 bond. This sequence is further evidenced by the geometric evolution in Figure 4B; in the transition state **TS2B'**_{C4-N3}, the C4-N3 bond significantly elongates to 2.135 Å, while the C2-S1 bond (1.864 Å) undergoes only negligible stretching compared to its length in **INT1B'**. This initial step leads to the formation of the ring-opening intermediate **INT2B'**, in which the C4-N3 bond is effectively cleaved (2.370 Å).

Following the formation of **INT2B'**, the completion of the hydrolysis process requires the further cleavage of the C2-S1 bond. Two potential mechanisms for this step were compared: direct bond cleavage and a concerted pathway assisted by proton (Hc) transfer from the C4-hydroxyl group (Figure 4A). The results indicate that the direct cleavage of the C2-S1 bond involves a relative barrier of 20.7 kcal/mol (relative energy of 30.8 kcal/mol at **TS3B'**_{C2-S1}). In contrast, the proton-assisted mechanism reduces the barrier to 19.0 kcal/mol (**TS3B'**_{Hc-S1}), identifying it as the more energetically favorable route for this stage. As shown in Figure 4B, **TS3B'**_{Hc-S1} exhibits typical concerted characteristics with highly coupled proton transfer and bond cleavage. In this transition state, the Hc proton transfers to the S1 atom, with the Hc-S1 distance shortening to 1.580 Å as the S-H bond forms. Simultaneously, the C2-S1 bond is significantly polarized and weakened by the proton transfer, with its length increasing to 2.286 Å. Subsequent to this transition state, the C2-S1 bond undergoes complete cleavage (increasing to 3.458 Å), leading to the formation of the ring-opening product CHBT and the corresponding byproducts.

Given the involvement of proton transfer in the C2-S1 bond cleavage, the potential for a water-mediated relay to further lower the energy barrier was also evaluated. Accordingly, an explicit water molecule from the active pocket was included in the QM region (model **INT1B''**) to determine if a water-mediated pathway would be more favorable. The resulting energy profile and key geometries are illustrated in Figure S7. In this model, the initial C4-N3 bond cleavage occurs with a barrier of 15.7 kcal/mol (**TS2B''**_{C4-N3}), forming intermediate **INT2B''**. However, the subsequent water-mediated proton relay for C2-S1 bond breaking requires overcoming a barrier of 26.6 kcal/mol (**TS3B''**_{Hc-S1}). This energy requirement is substantially higher than the 19.0 kcal/mol observed in the direct transfer pathway (**TS3B'**_{Hc-S1}).

As shown in Figure S8, the mediating water molecule in **INT2B''** is positioned 4.700 Å away from the Hc proton. This structural characteristic arises because the water molecule is sequestered within a stable hydrogen-bond

network formed with surrounding residues (Figure S8). The energy penalty required to reorganize this network for proton transfer accounts for the significantly increased barrier. In summary, due to the constraints of the localized hydrogen-bond environment, a water-mediated proton relay is not a favorable pathway for the ring-opening stage. Instead, the mechanism involving direct hydroxyl-proton transfer coupled with C2-S1 cleavage is identified as the dominant pathway for the final stage of hydrolysis.

Overall mechanism of LRE-catalyzed oLu hydrolysis

Based on the systematic evaluations described above, the complete optimal reaction pathway for LRE-catalyzed oLu hydrolysis was established. The overall potential energy surface and the structural evolution of the key stationary points are summarized in Figure 5. This hydrolysis process follows an “addition-stepwise ring-opening” catalytic mechanism, ultimately generating CHBT and the corresponding byproducts.

The catalytic cycle initiates from the **oLu + 2WAT + Asp** complex with an Asp111-assisted nucleophilic addition. During this stage, a water molecule attacks the C4 position of **oLu** while the adjacent Asp111 acts as a general base to abstract a proton. This concerted process, involving simultaneous proton transfer and C-O bond formation, proceeds via transition state **TS1B'** with an activation barrier of 14.3 kcal/mol. The addition reaction disrupts the planar conjugation of the thiazole ring and converts the C4 atom from sp² to sp³ hybridization, forming the addition intermediate **INT1B'**. The low activation barrier and the stability of the intermediate (3.4 kcal/mol) indicate that this initial step is energetically favorable.

Following addition, the system enters the ring-opening stage, which begins with the cleavage of C4-N3 bond. Starting from **INT1B'**, the system overcomes a barrier of 10.2 kcal/mol via transition state **TS2B'**_{C4-N3}. In this step, the C4-N3 bond significantly elongates and cleaves, allowing the system to relax into a second intermediate **INT2B'**. At this point, the partial ring opening at the nitrogen atom provides the necessary conformational conditions for the final cleavage.

The final stage of hydrolysis involves the cleavage of the C2-S1 bond. This process occurs through a concerted mechanism where a proton from the C4-hydroxyl group is directly transferred to the S1 atom. The resulting transition state, **TS3B'**_{Hc-S1}, represents the global energy maximum on the potential energy surface, with a relative energy of 32.5 kcal/mol. This indicates that the conversion from **INT2B'** to the final products requires overcoming a local barrier of 19.0 kcal/mol. Consequently, this

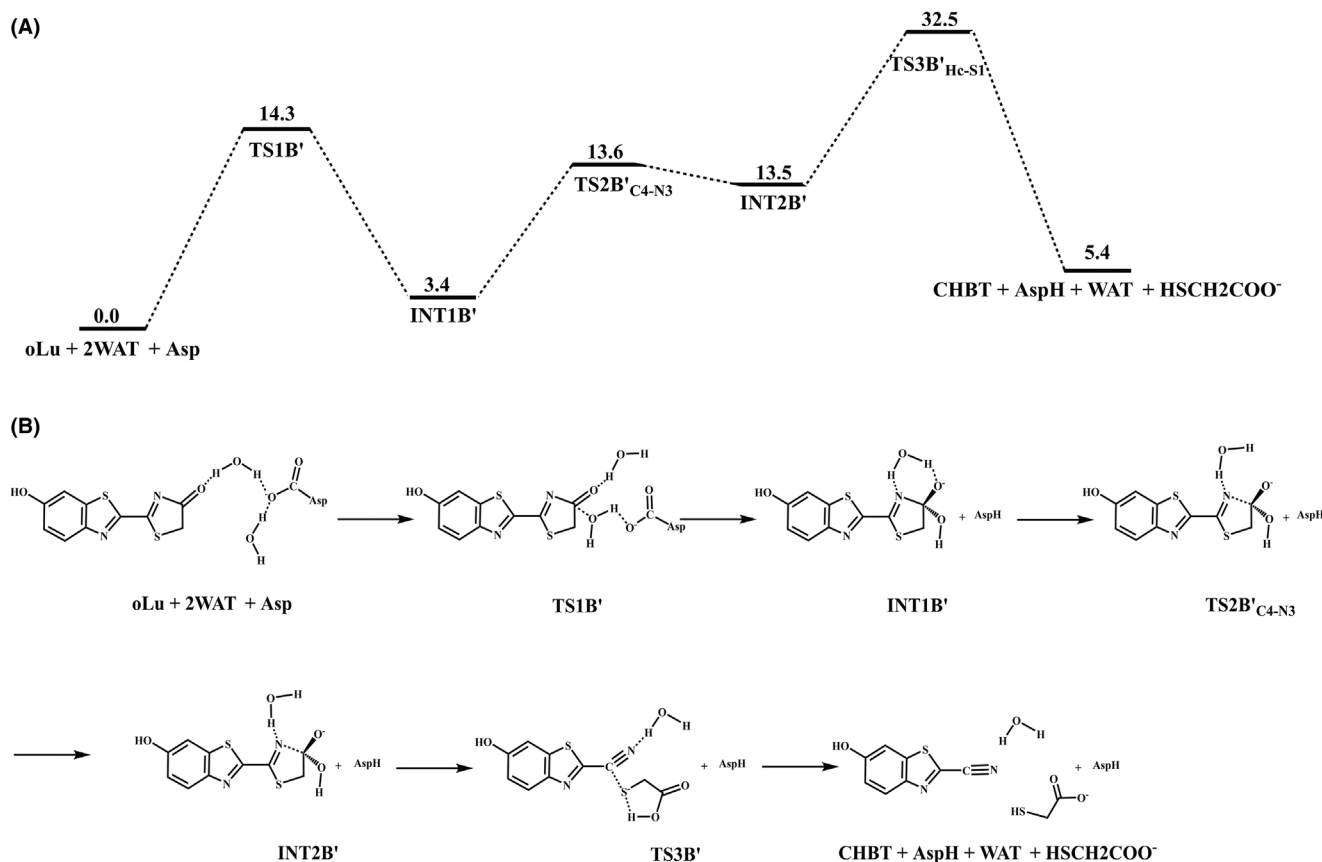


FIGURE 5 (A) Potential energy profile (ΔE , kcal/mol) and (B) schematic representation of structural changes for the optimal pathway of LRE-catalyzed **oLu** hydrolysis, calculated at the M06-2X/def2-TZVP//B3LYP-D3/6-31G*/MM level of theory.

proton-assisted C-S bond cleavage is identified as the rate-determining step of the entire LRE-catalyzed **oLu** hydrolysis process.

After crossing the rate-determining step **TS3B'**_{Hc-S1}, the system generates the final product complex (**CHBT + AspH + WAT + HSCH2COO⁻**). The relative energy of this product complex is 5.4 kcal/mol, indicating that the entire hydrolysis reaction appears as a slightly endergonic process under the computational model. However, in the real physiological microenvironment, this slight thermodynamic disadvantage may be compensated through the synergistic effects of multiple factors. On the one hand, the release of product CHBT from the LRE active pocket and the accompanying enzyme backbone conformational relaxation may provide certain thermodynamic compensation for the system, such as reducing the overall process disadvantage through entropic gains; on the other hand, this hydrolysis reaction is the initial step of the entire substrate regeneration pathway, and the primary product CHBT generated will subsequently undergo an exergonic condensation reaction with L-Cys that has a strong spontaneous tendency. The consumption of CHBT by this condensation reaction may reduce its accumulation in the system, thereby facilitating the hydrolysis step to proceed

toward the product direction. Overall, although this hydrolysis step shows slight thermodynamic disadvantage under the computational model, this disadvantage may be compensated to some extent under the synergistic effects of product release, enzyme conformational adjustment, and subsequent condensation reactions, making this process feasible in the complete luciferin regeneration pathway.

The molecular insights gained from this study provide testable hypotheses for future experimental validation. Specifically, our results identify Asp111 as the indispensable general base for the initial hydrolysis step. To verify this, site-directed mutagenesis could be employed; for instance, substituting Asp111 with residues lacking proton-accepting capability, such as alanine (Asp111Ala) or the non-basic asparagine (Asp111Asn), is predicted to significantly abolish or diminish the enzyme's catalytic competence by preventing the activation of the nucleophilic water molecule. Furthermore, the identified spatial arrangement (distances and orientations) of the Asp111–water–**oLu** catalytic triad identified here offers a structural blueprint for the de novo design and subsequent experimental characterization of biomimetic protein scaffolds. By grafting this specific three-dimensional

arrangement into artificial scaffolds, researchers can experimentally test whether this geometry alone is sufficient to reproduce the hydrolytic function. Given that recombinant LRE can be successfully expressed and purified in *E. coli* systems,¹⁵ and that the intermediate CHBT can be quantitatively monitored via mass spectrometry or kinetic assays,¹⁵ these experimental strategies are highly feasible and would provide crucial confirmation of the proposed catalytic mechanism.

CONCLUSION

This study systematically investigated the microscopic mechanism of LRE-catalyzed **oLu** hydrolysis within the firefly BL system using MD simulations and QM/MM calculations. The primary findings are summarized as follows:

First, MD simulations and binding free energy analysis characterized the LRE-**oLu** binding mode, identifying a strong binding affinity of -15.4 kcal/mol. The active site is characterized by the enrichment of nucleophilic water molecules near the substrate C4 atom and the strategic proximity of Asp111. These structural features demonstrate that Asp111 is ideally positioned to function as a general base for water activation.

Second, a comparison of two potential reaction pathways revealed that a direct, non-base-dependent nucleophilic addition is kinetically unfeasible due to a high energy barrier of 27.4 kcal/mol. In contrast, the Asp111-assisted mechanism—where Asp111 acts as a general base to deprotonate the water molecule—significantly reduces the activation barrier to 14.3 kcal/mol. This concerted addition step disrupts the thiazole ring's conjugation and leads to the formation of a stable sp^3 -hybridized intermediate **INT1B'**.

Third, the ring-opening mechanism of **INT1B'** was found to follow a stepwise sequence. The reaction initiates with the cleavage of the C4-N3 bond, which involves a low barrier of 10.2 kcal/mol, forming a partially ring-opened intermediate. Due to the constraints imposed by the localized hydrogen-bond network within the catalytic pocket, water-mediated proton transfer is energetically unfavorable. Consequently, the reaction proceeds via the direct intramolecular transfer of the C4-hydroxyl proton to the S1 atom, which facilitates the cleavage of the C2-S1 bond. This final cleavage step corresponds to the global energy maximum (32.5 kcal/mol) and is identified as the rate-determining step of the overall hydrolysis process.

In conclusion, these results provide a comprehensive molecular-level understanding of the LRE-catalyzed hydrolysis mechanism. The findings underscore the dual

role of the enzyme in providing both a precise geometric arrangement and a specific catalytic base (Asp111), ensuring the chemical feasibility of the luciferin regeneration pathway.

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DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the [Supporting Information](#) of this article.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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